



Designing a digital-twin based dashboard system for a flexible assembly line

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ABSTRACT

The current dashboard system is limited of coordination, flexibility and scalability. This study proposed a digital-twin based dashboard system. The digital twin components are divided into four, namely the operator, screw machine, industrial robot and automatic optical inspection in a flexible GPU card assembly line. This study designs a protocol enabling digital twins to interact and operate the system collaboratively and flexibly. In addition, this study presents a landscape of digital twin dashboards to demonstrate the relationship between the digital twins and the physical modules, providing a use case for the future digital-twin based dashboard design. The proposed system is implemented in a flexible GPU card assembly line and constructed by using JMobile to build a real-time dashboard system capable of scalability. The proposed dashboard framework for the GPU card assembly line offers several advantages over existing dashboards in terms of productivity, quality, and accident reduction. This study contributes to a scalable real-time dashboard system framework by incorporating human-machine collaboration, laying the foundation for advanced dashboard system development.

1. Introduction

With the rapid development of smart manufacturing, dashboard systems are becoming increasingly popular in enterprises and organizations. The dashboard system is a tool used to present and analyze data, including many systematic technologies, such as: management decision support, data analytics, human-machine interface (HMI), network and communication, and computing security (Wang et al., 2020), helping decision makers quickly understand production status, trends and key indicators (Vilarinho et al., 2017). However, existing dashboard systems have some limitations, such as lacks of coordination among devices, flexibility for exceptions, and scalability of functionalities, and human machine collaboration (HMC) capability (Reinking et al., 2020). Some studies have pointed out that it is extremely important to achieve the integration of 'hard' and 'soft' entities in dashboard systems, particularly human roles (Verdouw et al., 2021). A digital twin (DT)-based dashboard system serves as a critical tool for enabling human-machine collaboration by providing real-time monitoring, predictive analytics, remote access, simulation capabilities, and supporting continuous improvement efforts for machine and humans. By leveraging the strengths of both humans and machines through modeling machine and

humans as digital twins, manufacturing systems can enhance operational efficiency, effectiveness, and adaptability.

Several studies have pointed out that in DT technology, agents can be used to simulate various roles and behaviors in the real world, thereby achieving comprehensive simulation and analysis of the system, and real-time adjustments can be made according to changes in the environment (Wang et al., 2024). Through DT technology (Wang et al., 2021; Kumbhar et al. 2023), the behavior, rules, status and logic of physical objects (such as personnel, products, equipment, production lines, and environment) can be modeled in virtual space, and the models can communicate with each other (Chen & Huang, 2020), and further optimization solutions can be proposed (Wang & Luo, 2021). In addition, equipment information can also be made transparent to realize remote display, control and maintenance, so that managers can make accurate decisions to improve efficiency and safety (He et al., 2019).

Nevertheless, DT technology remains a gap in communication capabilities (Wang et al., 2020). The implementation of a dashboard system with quality communication capabilities enables seamless data exchange and information sharing between different manufacturing machines and entities (i.e., physical part/agent of DTs), and can instantly respond to changes in production conditions, to increase the

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coordination, flexibility, and scalability of intelligent manufacturing systems. The design and development of dashboard systems based on DT technology is still in its infancy, with promising research and development potential.

The HMC emphasizes the effective cooperation between humans and machines. The introduction of HMC into the dashboard system can make the system more flexible to meet the needs of users and improve the user's work efficiency and accuracy (Salunkhe et al., 2023; Angleraud et al., 2024). In addition, as the complexity of the work environment increases, especially tasks that require real-time monitoring of personnel status monitoring (Kerner et al., 2021), such as personnel fatigue and personnel stress, the adoption of dashboard systems considering human statuses becomes more important. Monitoring personnel status is also a crucial task, especially in work environments that need to ensure personnel safety, improve work efficiency, and respond to emergencies (Lin & Lukodono, 2022).

However, traditional monitoring systems only provide data visualization which is lack of an in-depth understanding of personnel status (such as fatigue and stress) and immediate feedback, making it impossible for managers to discover potential safety risks promptly and make rapid response measures. At the same time, the lack of personnel status data prevents managers from adjusting work arrangements and staffing according to actual conditions, which may also affect work efficiency.

This research aims to develop a DT-based dashboard system and use it in HMC settings. First, through literature and industry surveys, the gap in the application of DT technology in dashboard systems was determined, and a corresponding design prototype was proposed to fill the gaps in the technical application of existing dashboard systems. It strengthens the integration of heterogeneous platforms. Secondly, a simple and efficient communication protocol is also designed to implement the proposed DT-based dashboard system in the context of a flexible GPU card assembly line. The enhanced role of communication functions in system coordination, flexibility and scalability. Thirdly, it provides personalized HMC personnel physiological status monitoring to make up for the lack of current dashboard monitoring functions. Through this research, a highly integrated, intelligent, flexible and interactive dashboard system is proposed.

The remainder of this paper is organized as follows. Section 2 presents a literature review on dashboards, DT technology, and personnel fatigue and stress. Section 3 presents the framework of the dashboard system based on the DT, and the communication protocol of the DT dashboard system. Section 4 provides an implementation case of the dashboard system. Finally, Section 5 provides the result.

2. Literature review

2.1. Intelligent agent and digital twin technology

"Intelligent agent" is usually used to indicate the names of programs with different characteristics and functions, and has four characteristics, namely autonomy, social ability, initiative and responsiveness. Each individual entity is represented by an intelligent agent and can be run dynamically at runtime. The agent system is entity dynamic, which can divide complex tasks, assign tasks to the most suitable agents, and automatically arrange the agents. The intelligent agent technology has naturally evolved into multi-agent system (MAS) (Amato et al., 2021) with six core attributes: distribution, adaptability, flexibility, scalability, leanness, and resiliency (Liu et al., 2020). According to customer requirements of product orders, an MAS for manufacturing control can select the machine or resources to process it and achieve intelligent production and supply chain management (Ivanov & Dolgui, 2021).

DT technology is an emerging technology derived from intelligent agent technology, and integrates the physical and digital world (Kamble et al., 2022). It provides comprehensive physical and functional descriptions of compartments, products, or systems, and can be expanded independently and updated automatically (Schleich et al., 2017). DTs

represent physical devices or components by integrating data and simulation, and this technology has both static and dynamic information. The static information includes geometric sizes, lists of materials, and procedures, whereas the dynamic information includes information on the product life cycle that changes over time (Haag & Anderl, 2018). Through a large amount of information from various devices, it can be continuously updated and visualized to identify the equipment status of current and future (Mathupriya et al., 2020). At present, the implementation of DTs mostly revolves around data and communication interfaces, emphasizing that the communication between DT models requires more effective and safer communication protocols. (Xia et al., 2020).

The basic architecture of DT is composed of three core elements, namely physical space, virtual space and connection (Wang et al., 2021). The three core DT framework is only a preliminary form, and proposed five elements of digital space, virtual space, connection, data and service to form a complete DT framework (Tao et al., 2018a).

The five elements of the DT framework are elaborated as follows. (i) digital space: refers to the environment where the physical entity exists and operates. It encompasses the physical infrastructure, machinery, sensors, actuators, and other components that constitute the real-world system. (ii) virtual space: represents the digital counterpart or replica of the physical entity in the digital realm. This virtual model includes all the relevant information, characteristics, behaviors, and functionalities of the physical entity, enabling simulation, analysis, and experimentation. (iii) connection: denotes the seamless link or interface between the digital and virtual spaces, facilitating bidirectional data exchange, communication, and synchronization between them. This connection ensures that changes in the physical entity are reflected in the virtual model and vice versa, maintaining consistency and accuracy. (iv) data: encompasses the vast amount of information collected from the physical entity and generated within the virtual model. This includes sensor data, performance metrics, operational parameters, environmental conditions, and any other relevant data sources that contribute to the understanding and optimization of the system. And (v) service: represents the functionalities and capabilities provided by the digital twin framework to support various applications, such as monitoring, control, optimization, prediction, and decision-making. These services leverage the data and insights derived from the digital and virtual spaces to enable intelligent operations and management of the physical entity.

It can be used both to predict the result of certain purposes and to predict product and service behaviors (Tao et al., 2018b). DT has proved to be a way to integrate the physical world and the virtual world of manufacturing, and have three levels: monitor, predict, and decide (Fonseca et al., 2022). It can building virtual models of physical equipment, production lines, factories, and people, as well as related data repositories, and record its operation process and decision-making information, further use of DT system monitoring, control and simulation of physical real status (Liu et al., 2023).

In summary, intelligent agent technology, characterized by autonomy, social ability, initiative, and responsiveness, has evolved into MAS with enhanced adaptability, flexibility, scalability, leanness, and resiliency, effectively managing production and supply chains by assigning tasks to suitable agents. DT technology, building on intelligent agent principles, integrates physical and digital realms to provide comprehensive descriptions of physical entities, leveraging both static and dynamic information for real-time updates and visualizations. A robust DT framework, consisting of digital space, virtual space, connection, data, and service, facilitates seamless interaction between physical and virtual models, supporting monitoring, control, optimization, prediction, and decision-making processes, thereby enhancing manufacturing efficiency and decision-making capabilities. In this study, the exiting literatures contribute to highlighting the integration of MAS and DT technologies, demonstrating their potential to enhance smart manufacturing by connecting the physical and digital domains and improving system monitoring, prediction, and decision-making capabilities.

2.2. Modern dashboard systems

Dashboard system is a visual performance management tool that visualizes information about important of organizational goals, such as requirements, direction, and conducts action guidance, as well as support and improve on-line decision making (Bachechi & Rollo, 2022). With the development of smart manufacturing, the need to dashboard systems becomes critical and challenging (Vahedi et al., 2022). Dashboards have different definitions, such as decision theater (Boukherroub et al., 2018), central control room (Rustambekovich et al., 2017), war room (Kumsap et al., 2018), interactive whiteboard (Alvarez et al., 2013), and digital signage (Oh et al., 2015).

The dashboard provides different management functions. For example, it can provide a storage device to monitor, to manage energy operation, to control devices, to manage orders and machines, to enhance human insight and long-term/short-term decision making, and to control scenes and databases. Further, it improves management team productivity and is highly adaptable to the environment (Franceschi et al., 2022).

DT based digitalization technology provides modeling, data exploration, and data analysis to visualize data and make decisions in a new way, resulting in the emergence of modern dashboard systems (Boukherroub et al., 2018) in a variety of industries. More related literatures (Bachechi et al., 2022; Deryabin et al., 2020; Fu et al., 2016; Grandi et al., 2022; Greif et al., 2020; Malik and Brem, 2021; O'Dwyer et al., 2020; Pan et al., 2020; Salunkhe et al., 2023; Tao et al., 2018a; Tao et al., 2018b; Weckx et al., 2022; Xie et al., 2021; Zhang et al., 2021) are shown in Online Appendix 1.

In summary, the evolution of dashboard systems has significantly enhanced visual performance management, particularly in smart manufacturing. These systems facilitate the visualization of critical organizational goals and support real-time decision-making. Modern dashboards, provide diverse management functions including monitoring, energy management, device control, and order management. They enhance human insight, productivity, and adaptability to the environment. The integration of DT-based digitalization technology has further advanced dashboard systems by enabling sophisticated modeling, data exploration, and analysis, thereby improving data visualization and decision-making capabilities across various industries. These related literatures help highlight how to improve organizational goal visualization, support online decision-making, and increase operational efficiency in complex manufacturing environments.

2.3. Personnel fatigue and stress

Personnel fatigue refers to the physical and psychological fatigue caused by long hours of work or other activities. This state may have a negative impact on an individual's physical and mental health, while also jeopardizing work productivity and safety. Especially in occupational fields that require long periods of concentration and high concentration, such as transportation, manufacturing, and health care.

The measurement of personnel fatigue varies depending on the research purpose and methods, and different classification systems may be used in different literature, such as: self-report, questionnaire feedback, model building, physiological indicators (such as heart rate, brain radio waves) and psychological indicators (such as subjective reports and attention levels) (Umer et al., 2022). Physiological indicators are most common method, which uses a multi-level classification system, usually three to five levels, to describe the degree of fatigue (Antwi-Afari et al., 2023), such as mild fatigue, moderate fatigue, severe fatigue, extreme fatigue, dangerous state. The purpose of personnel fatigue classification is to more comprehensively understand and classify an individual's fatigue level during long-term work or activities, which helps to determine an individual's fatigue level more accurately.

Moreover, personnel work stress is usually defined as the physiological and psychological responses produced by stressors, skill or job

mismatches that individuals face in work or life. Study indicates that pointed out that excessive work stress harms an individual's physical and mental health, and will also be reflected in employees' work performance and work efficiency, thereby affecting the overall performance of the organization (Mansour, 2016). Based on the type and degree of stress sources, some studies have proposed a classification of work stress levels, dividing work stress into three to four levels (Arsalan & Majid, 2021), which divided levels: mild, moderate, high and extremely high, and help organizations more accurately understand the level of stress faced by their personnel and thus adopt corresponding response strategies.

To more systematically assess and compare stress levels in different working environments, much literature has proposed various tools and methods, including stress questionnaires, physiological indicator monitoring, etc. (Nwaogu & Chan, 2021) to understand and evaluate this phenomenon more comprehensively. Physiological indicator monitoring is one of the most common methods. Many studies use physiological indicators to objectively assess an individual's stress level, including blood pressure, cardiac response, electrodermal activity (EDA), Electroencephalogram (EEG) and electromyographic (EMG). (Willemse & Van Erp, 2019), changes in these physiological responses can reflect the physiological changes of individuals under stress. For example, increases in heart rate and skin conductance are often associated with rising stress levels (Hanshans et al., 2023).

In summary, personnel fatigue and work stress are critical factors affecting individuals in various occupational fields, including transportation, manufacturing, and healthcare. Fatigue, stemming from prolonged physical and psychological exertion, poses risks to both health and productivity. It is assessed through diverse methods like self-reports, physiological indicators, and psychological assessments, aiming to accurately gauge and classify fatigue levels. Similarly, work stress arises from mismatches between job demands and individual capabilities, impacting physical and mental well-being, as well as organizational performance. Classifying stress levels helps organizations implement targeted interventions to mitigate its effects. In this study, these addressed literatures help to highlight the importance and impact of stress and fatigue rating systems in the work environment. Through a variety of classification systems and measurement methods, such as physiological indicators and psychological assessments, researchers are able to more comprehensively assess an individual's fatigue and stress levels in order to increase productivity, promote employee health and safety, and improve the overall performance of the organization.

2.4. Summary

Based on the related literatures, we can draw some findings that contribute to this study, and at the same time identify some gaps in the research. It is worthy to note that DTs can use intelligent agents to build virtual models of entities, including physical equipment, production lines, factories, and people, as well as related data repositories (Liu et al., 2023; Didden et al., 2024), to simulate the real-time manufacturing environment. Agent technology satisfies the need to construct DTs by dividing complex tasks, assigning tasks to the most appropriate agents, and automatically arranging agents to simulate the behavior of individuals in the environment (AboElHassan et al., 2023). However, there are still some gaps in currently practical dashboard systems and theoretical modelling when it comes to integrating DT and agent technologies (Wang et al., 2020; Wang et al., 2021; Franceschi et al., 2022).

First of all, most of the current research literature discusses using agent systems to construct DT models of equipment and capture equipment operating status. However, there are still limitations in displaying personnel fatigue (Lin & Lukodono, 2022; Angleraud et al., 2024). Although DT and agent technologies can simulate human body movements and reactions, there is a lack of effective indicators to objectively assess personnel fatigue. This results in the dashboard being unable to provide comprehensive fatigue analysis, limiting insights into

work performance and employee status.

Secondly, dashboard systems also have shortcomings in displaying a person's stress levels. DTs combined with agent technology can simulate human responses under different stress conditions, but current dashboards fail to effectively capture these subtle changes. Lack of accurate monitoring of stress factors may result in the inability to identify employee work stress problems early, which in turn affects productivity and efficiency (Antwi-Afari et al., 2023).

Finally, the existing dashboard systems have not fully integrated data from DTs and agent technology, resulting in incomplete information presentation (Liu et al., 2023). Due to the huge amount of data generated by multiple systems, current dashboards cannot fully utilize this information to provide deeper analysis and instant feedback (Ding et al., 2019). This makes decision-making on improving the work environment and employee performance relatively difficult.

Overall, existing dashboard systems and related theories need to be further developed to overcome the abovementioned gaps to enable more comprehensive and accurate personnel status monitoring and management. Therefore, it is promising to further develop a HMC dashboard system framework based on DTs by combining the advantages of the DT

concept with agent technology and considering the shortcomings in sensing and monitoring the physiological and psychological states of personnel.

3. Proposed DT based dashboard system – a flexible GPU card assembly line

The DT-based dashboard system offers more than just monitoring capabilities—it also includes scheduling and control functions. By integrating physical equipment with virtual models, the system can continuously monitor the status of critical operations and equipment along the assembly line in real-time. Additionally, it automatically adjusts the schedule and operation of the assembly line based on real-time data and equipment status, providing seamless coordination between virtual and physical components while reporting these adjustments to the physical equipment interface.

The manufacturing environment landscape for DT dashboards is shown in Fig. 1. It is divided into three layers. The first layer involves building physical production line equipment, integrating equipment from different platform systems, and establishing a flexible GPU card

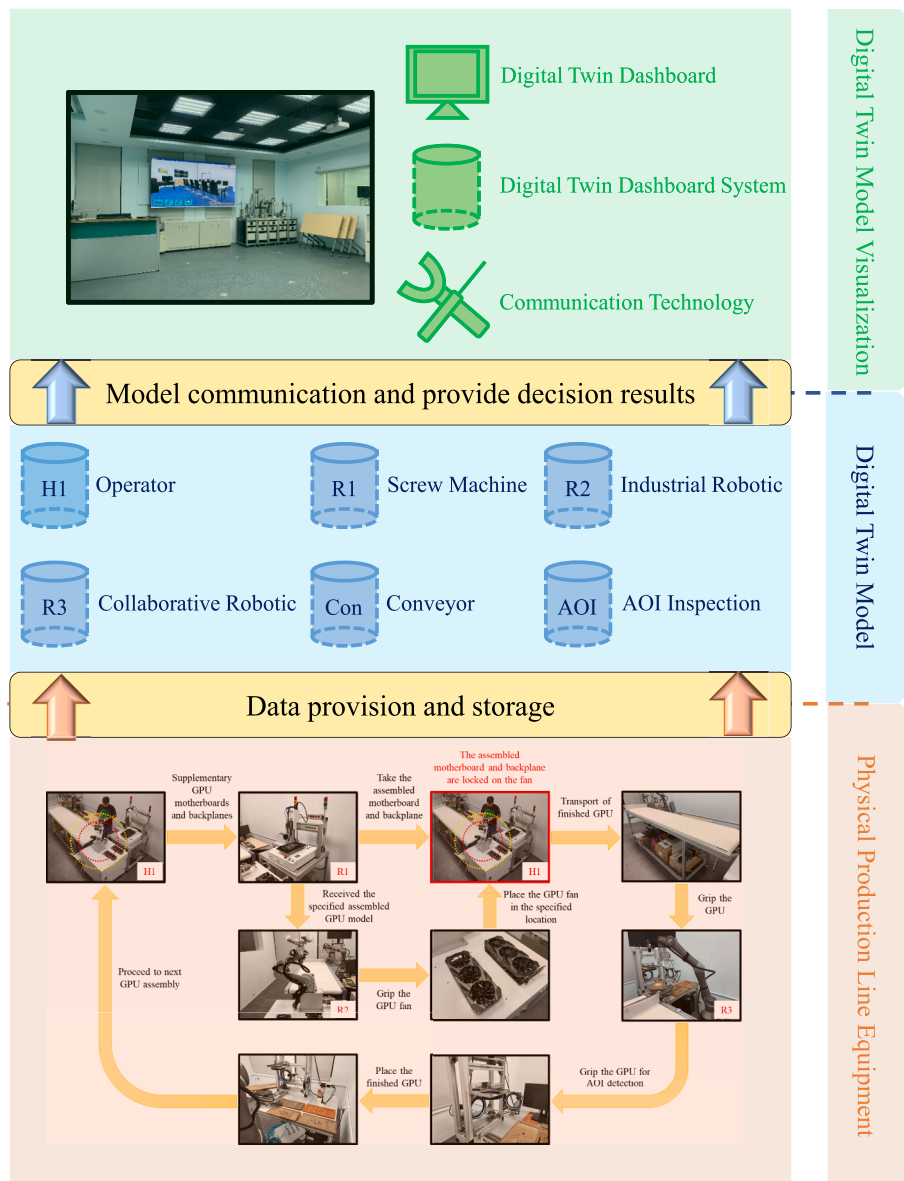


Fig. 1. Digital-twin based dashboard system – an overview.

assembly line scenario. In the second layer, DT models of physical equipment are constructed, including operators, screw machines, industrial robotic arms, conveyor, collaborative robotic arms, and automatic optical inspection (AOI) station, and assigned work tasks. In the third layer, the dashboard hardware and communication technology are integrated.

3.1. Developing DT components

A flexible GPU card assembly line installed in the smart manufacturing laboratory at National Taiwan University of Science and Technology is set as the target system of the study. The line conducts the final assembly of GPU cards. Each physical device has a corresponding virtual agent, including the screw machine (R1), industrial robotic arm (R2), collaborative robotic arm (R3), operator (H1), conveyor belt (Con), and AOI. The production control process is shown in the lower part of Fig. 1.

A GPU card assembly line involves several key operations and equipment. The following is a functional description of each operation and device (the details refer to Online Appendix 2): The operator (H1) is responsible for picking up and replenishing information and is also responsible for locking the GPU card motherboard and fan. The screw machine (R1) is an automatic screw-locking machine. Its main responsibility is to lock the protective bottom plate of the motherboard and GPU card. The industrial robot (R2) is a six-axis industrial robot. It is responsible for clamping the GPU card fan in a fixed position and moving and fixing it in the corresponding position for H1 to assemble. The collaborative robot (R3) is a collaborative robot with a vision. It is responsible for taking the finished GPU card to the AOI for inspection. The conveyor belt (Con) is a conveyor belt that transports finished GPU cards. It is responsible for transporting finished GPU cards from R2 to R3. Automatic optical inspection (AOI) is an automatic optical inspection module responsible for GPU card defect detection.

The DT components on the flexible GPU card assembly line is constructed using Python software. Python real-time programs are created using two python components, namely RPyC and Socket (Fig. 2). The RPyC thread, a Python module for distributed computing, will trigger a program in each assigned PC to create each associated proxy, while the

socket thread, the module responsible for handling communication between agents, will create a server interface that continuously listens and passes messages to other clients by following Modbus TCP/IP. Each proxy created by RPyC has its own client socket and can communicate with the server and other agents.

On the flexible GPU card assembly line, the interaction between the physical device and the virtual model is a dynamic and intimate process. As the core of the control equipment, the PLC is responsible for monitoring and controlling various operations on the assembly line. The virtual model simulates and controls the operations on the physical assembly line by communicating with the PLC.

Accordingly, the proposed DT-based dashboard system not only has monitoring related functions, but also scheduling and control functions. Through the interaction between physical equipment and virtual models, the system can monitor the status of various operations and equipment on the assembly line in real-time. At the same time, the system can also automatically adjust the schedule and operation of the assembly line based on real-time data and equipment status.

3.2. Proposed protocol enabling communication among DTs

Contract Net Protocol (CNP) is a protocol used for negotiation and task allocation in MAS (Didden et al., 2024). CNP controls the cooperation of two roles related to task execution through an interactive auction process (Wang et al., 2004; Wang et al., 2021). The agent can be a manager responsible for monitoring and proving the results of the task, or a contractor responsible for performing the task. Based on CNP, this study designed a robot communication protocol. Each node communicates with other nodes so that they can agree on rules for resource allocation and access to resources. This protocol guides the interaction between auction clients (C nodes) by allocating tasks to schedulers (Human + R1, R2, R3, Con and AOI). In this round of auction, there is no communication between dispatchers. The communication exists between the client and the dispatcher, as shown in Fig. 3.

The CNP process is divided into four steps, including job announcement, local planning and submission, schedule plan evaluation and schedule plan confirmation. By implementing the CNP based scheduling process, this study has considered three types of events (the

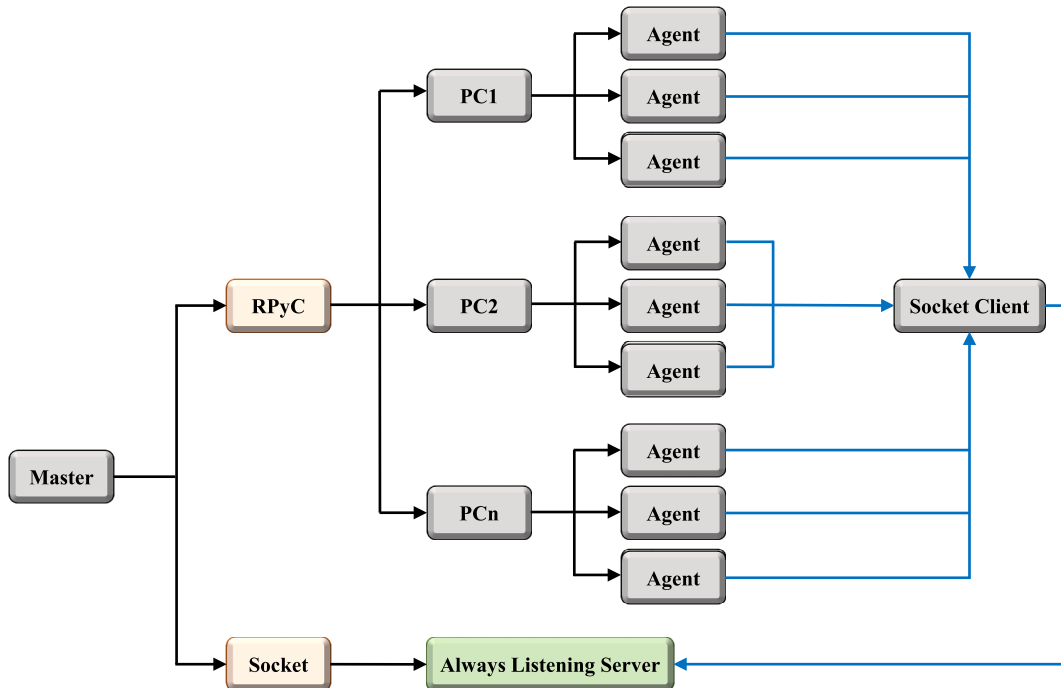


Fig. 2. Agent creation architecture (Master program).

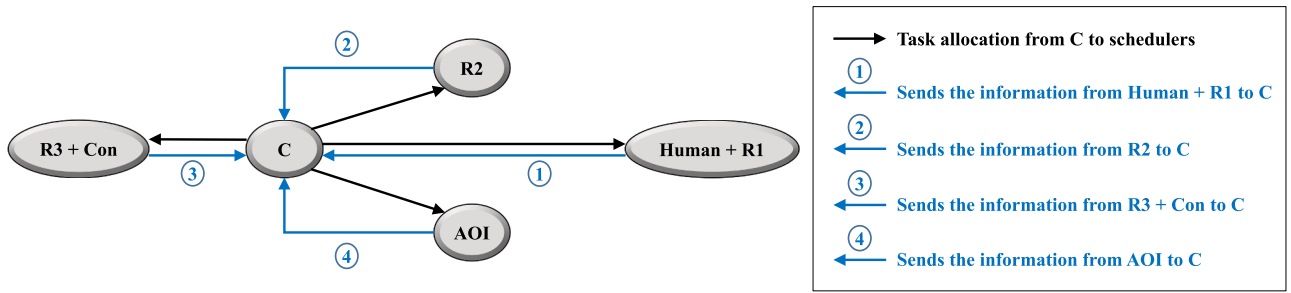
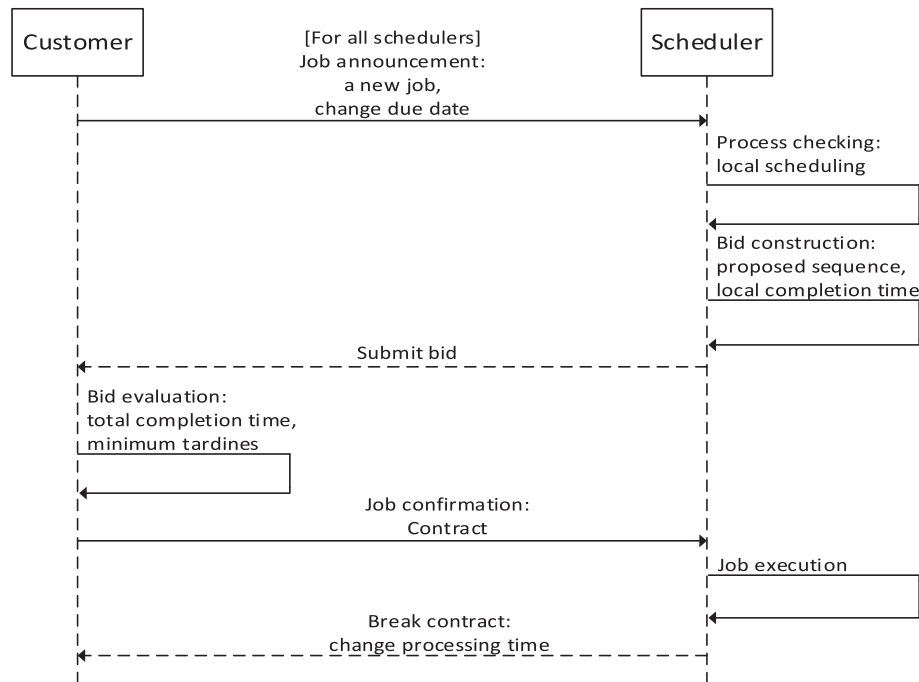


Fig. 3. Contract network protocol communication in the study.

details refer to Online [Appendix 3](#)): (i) arrival of new job, (ii) change due date and (iii) change processing time to adapt to the dynamic production environment. In CNP, the interaction between the customer agent acting as the manager and the dispatcher agent acting as the contractor, which are eight programs, as shown in Fig. 4. The client broadcasts job announcements to all schedulers triggered by new jobs or changed due dates. The scheduler receives messages from the client and checks its capabilities based on local information (including jobs in the queue, processing time, and waiting time) to process the local scheduling algorithm. The local scheduling algorithm (Veeramani & Wang, 2006) is applied to arrange a series of jobs, as well as the completion time of each

job, and propose to the customer. In addition, all available schedulers will bid to the customer, and the customer will evaluate the bid to select the overall best total completion time. The customer then sends the contract and confirmed work to the dispatcher. The dispatcher accepts the contract and performs the work. However, when the scheduler interrupts the contract due to a change in processing time, it will trigger a new event, and the scheduler will start sending a message to the customer, and then the customer will conduct another auction for the work on the contract.

In the example of the GPU card assembly line, this process can be applied to multiple operations, such as the use of R1, the operation of R2



Step1. Customer broadcast a new job or a change due date to the schedulers.

Step2. Scheduler does the local scheduling based on customer's information (ongoing jobs and processing time)

Step3. Scheduler produces jobs sequence and resulted the completion time

Step4. Submit the bid to the customer

Step5. Customer evaluates bids from schedulers and choose the best sequence and completion time

Step6. Customer send the contract to the scheduler

Step7. Scheduler execute the contract

Step8. The contract is broken by scheduler when the processing time is change and cannot fulfill the proposed completion time

Fig. 4. The interaction diagram between the client and the dispatcher in the CNP.

or R3, etc., to achieve efficient collaboration and operation of the production line. By implementing CNP in a digital twin environment, the efficiency and flexibility of task allocation and collaboration can be significantly improved. For example, when there is a batch of new GPU card motherboards that need to lock the backplane, the auction client will send a contract invitation (Call for Proposals, CFP) to the relevant R1 and H1. The R1 and H1 will evaluate whether they can accept the task based on the current working status and capabilities, and submit a bid (Proposal), which includes information such as the estimated time to complete the task and required resources. The auction client will select the most suitable agent based on the bids received and send a contract award notice (Award Contract) to the selected agent. After receiving the notification, the R1 and H1 began to operate locking the base plate, and continued to report the task progress to the manager agent. It can not only improve the collaboration efficiency of the production line, but also achieve flexible adjustment and optimal configuration of the production

process. Even if emergencies occur during the production process, the system can respond and adjust quickly to ensure production continuity and efficiency.

The protocol is implemented using python software and operates using a graphical user interface (GUI). The Master program splits itself by using multi-threading. The server created by the Master program will continuously listen and try to send messages to any agent. Each agent also has its own GUI (the details refer to Online [Appendix 4](#)).

3.3. Proposed advanced interface framework design

Recent literature defines DT as three elements: physical space, digital space, and the connection between both ([Wang et al., 2021](#)). In addition to these three types, the DT framework contains two elements: data and services ([Tao et al., 2018a](#)). Therefore, this study integrates the framework proposed in recent literature and proposes a DT design framework

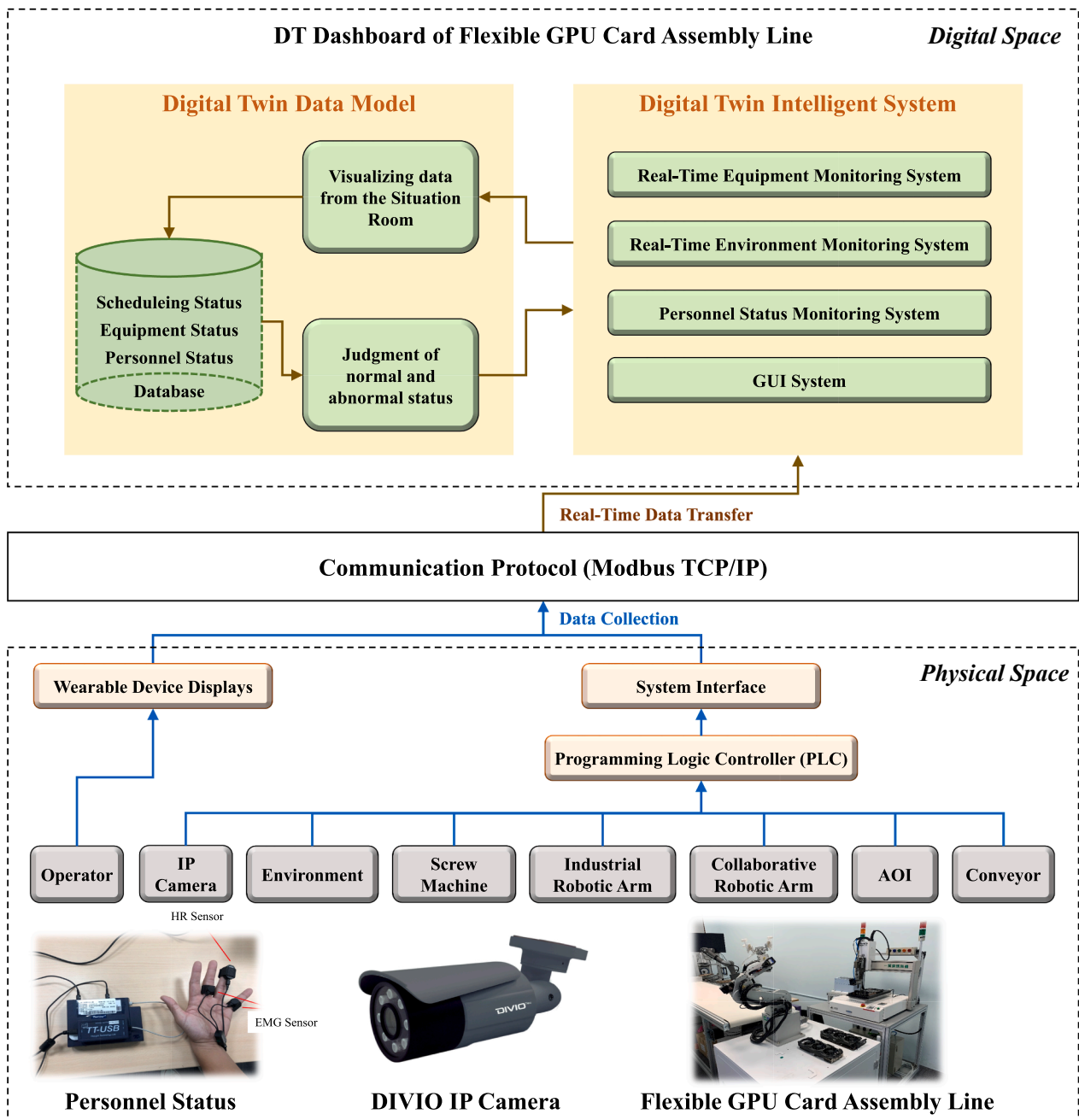


Fig. 5. Proposed DT design framework for flexible GPU card assembly line.

for monitoring people and machines in the HMC context, which is divided into three parts: data capture, DT intelligent system and DT data Model, as shown in Fig. 5.

In this study, the real-time operating data of each device is captured from the physical space, accessed and displayed on its system interface. On each equipment system interface, various statuses of the machine will be displayed, such as equipment coordinates, moving speed, clamping status, abnormal status, voltage, environment temperature, environment humidity, environment carbon dioxide concentration, PM2.5 value, etc. The operator's real-time data is captured through wearable devices, such as heart rate and skin conductivity. Each system interface is also connected to the DT dashboard for real-time monitoring. The details of equipment parameters and agent settings refer to Online Appendix 5.

In terms of network configuration, there are currently many studies using different methods, such as AutomationML, MTConnect, and OPC UA (Ding et al., 2019). In view of the IoT requirement of the flexible GPU card assembly line, this study uses ethernet for network configuration. The DT dashboard uses the second type of industrial network Modbus TCP/IP communication protocol (refer to Fig. 6).

The proposed protocol is implemented to facilitate seamless communication among various equipment, ensuring real-time data exchange, flexibility, and reliability. By utilizing the Modbus TCP/IP communication protocol, the proposed DT framework can effectively interact with the physical components of the flexible GPU card assembly line, enabling scalable integration and efficient data transmission.

The establishment of the DT model is achieved by deploying the Modbus TCP/IP communication protocol, enabling the realization of objectives of the six agents (H1, R1, R2, R3, Con, AOI). Consequently, such agent-based modeling and implementation contributes to robustness and adaptability (Tchana et al., 2019).

This study is conducted in five stages in developing DT intelligent systems. In the first stage, observe and organize the physical equipment operation area and various equipment issues, and discuss with engineers, such as system operation, equipment operation, parameter adjustment, and scenario design. In the second stage, understand the various requirements for DT dashboard design, such as the number and type of interfaces and the parameters required to be displayed. In the

third stage, refer to the JMobile software development template (Exor, 2023) to facilitate the subsequent development of the DT dashboard. In the fourth stage, the DT dashboard is personalized designed and the source of the required digital data is ensured. For example, the parameters set by heterogeneous platforms have different sources and methods, so the source location of each parameter must be ensured. Finally, test the equipment connection between the DT intelligent system and the physical equipment operation area for implementation and evaluation, such as: whether the equipment is connected normally, confirming the parameter source, and whether the program judgment thread is normal.

The proposed DT based dashboard system consists of four functions, namely real-time equipment monitoring, real-time environment monitoring, personnel status monitoring, and GUI module. The real-time equipment monitoring can display real-time parameters of the equipment and assist the operator in judging the machine's running trajectory, operating status, voltage, temperature, production category. The environmental monitoring helps operators understand the health of their work environment. The personnel status monitoring can instantly watch the operator's physiological and psychological status, such as: heart rate (fatigue), skin conductivity (stress level). The GUI system assists operators in production line scheduling.

The DT data model explains how the data in the DT intelligent systems operates (Ding et al., 2019). This model has three elements, including real-time data display on DT dashboards, data integration on server database, and database on heterogeneous platforms for each device, as shown in Fig. 7. In this study, each device has a data platform, and the data and parameters of each device are integrated into the server database through Modbus TCP/IP communication. The DT dashboard will compile and judge the data or parameters obtained from the server database through Javascript, and then display them on the DT dashboard.

4. Implementation of the digital-twin based dashboard system

The proposed dashboard system is divided into two areas, which are the physical virtual model monitoring area, and the physical equipment operation area (Fig. 8). (i) virtual model monitoring area: this area is

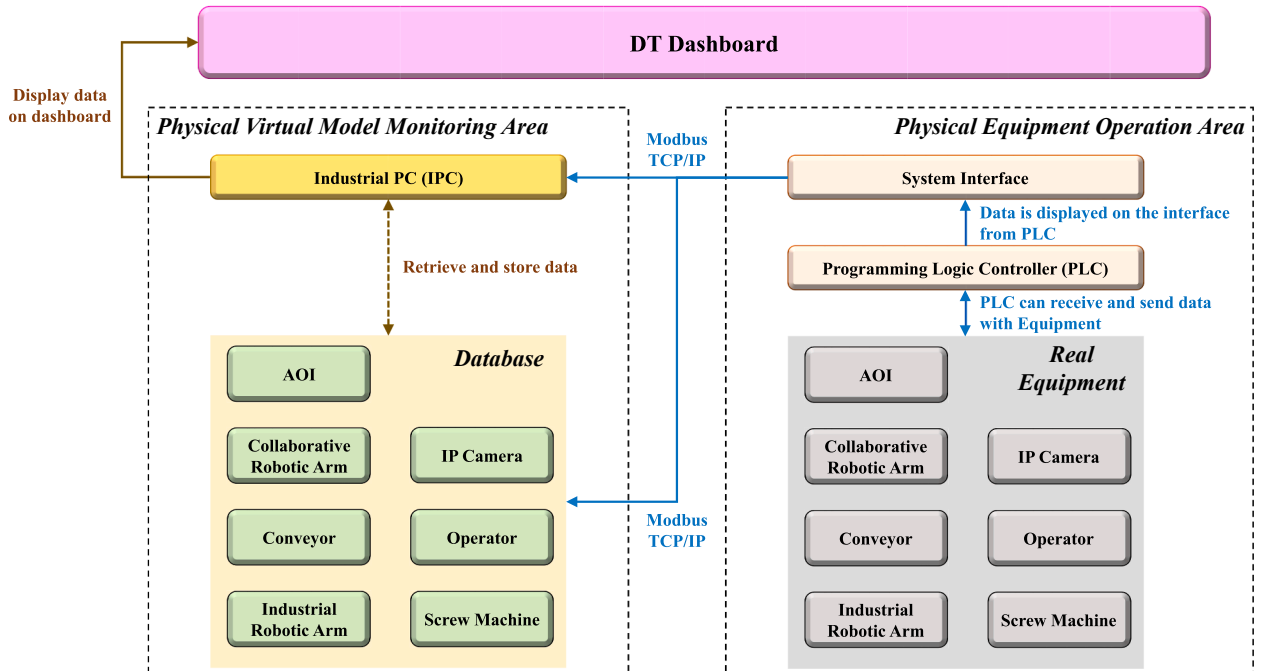


Fig. 6. Data capture model.

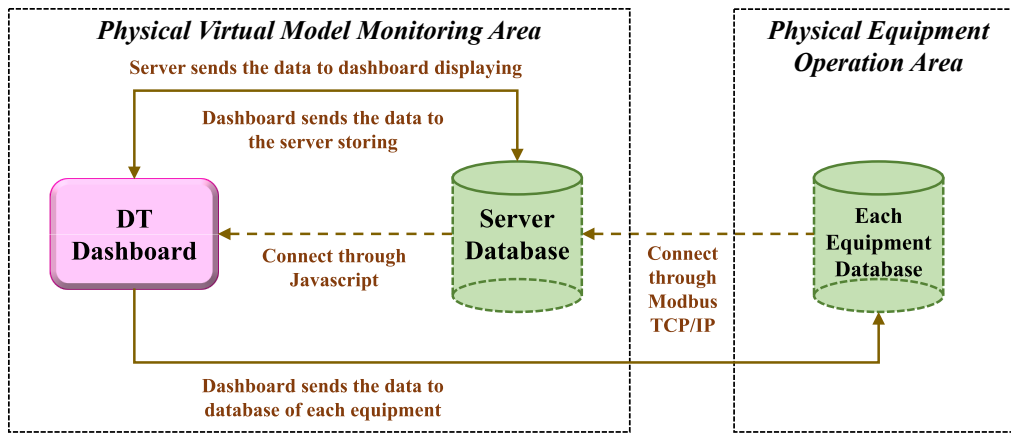


Fig. 7. The proposed operation method of DT dashboard database.

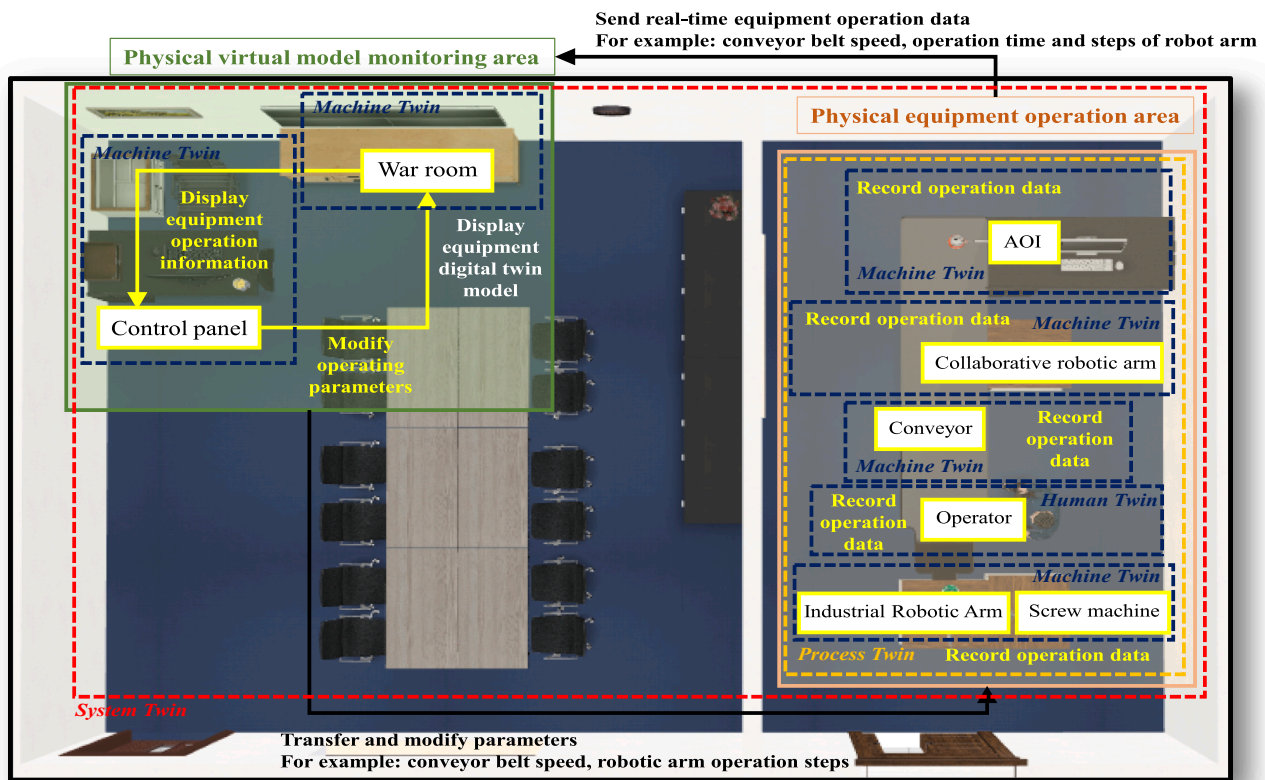


Fig. 8. DT technology implementation.

divided into two areas, dashboard system panel and discussion area. The dashboard system panel records and displays the information of the device and status of the actual operation of the device. The discussion area is available for managers to hold meetings and discussions. (ii) physical equipment operation area: this area is divided into six areas, conveyor, industrial Robotic Arm, collaborative robotic arm, screw machine, AOI and operator. The conveyor can provide demonstrations for carrying and transporting products. The robotic arm area can provide demonstrations of robotic procedures and physical operations. The screw machine can provide the process of screwing the unassembled parts (base plate) and interactive demonstration with the operator. The AOI can provide a demonstration of product defect rate detection. The operator can provide a demonstration of processing materials and processing defective parts.

Through utilizing PLC, multiple agents (H1, R1, R2, R3, Con, AOI)

are connected to the physical equipment. Subsequently, the captured real-time data are transmitted to the control panel (main console) using Modbus TCP/IP communication protocol. A two-way communication is constructed and synchronized between the control panel and the war room. When any abnormality is found, the remote war room will send signals to the control panel to adjust receipts/parameters to the corresponding physical equipment.

To implement the dashboard system based on DTs, this study adopts JMobile (2022), which is an efficient HMI development software and can run on PC platforms to support PLC for multiple agents. JMobile provides a comprehensive suite of functionalities essential for building sophisticated dashboard systems, including connection, device management, process management and data visualization, especially suitable for machines with IoT functionality, making it easily to integrate DTs into manufacturing environments. JMobile connects to the PLC, and

create functions such as Trend, Alarm, Schedule, Recipe, etc. At the same time, it supports remote monitoring, so that managers can view the real-time dynamics of the equipment in distance.

The DT dashboard has eight functional options, including environmental monitoring, industrial robot arm monitoring, collaborative robot arm monitoring, AOI monitoring, HRC field monitoring, conveyor monitoring, IP camera monitoring and screw machine monitoring, as shown in Fig. 9. Detailed interface design refers to Online Appendix 6.

The real-time monitoring system can monitor the temperature and humidity changes in the field, and can also provide various data acquisition modules. Through the JMobile communication integration platform, the data of multiple heterogeneous platforms is integrated, so that managers can instantly grasp the information of different types and brands of machines and equipment in the current factory, such as the equipment signals of normal operation, standby and malfunction, so as to achieve transparency of equipment information, as shown in Fig. 10.

The robot arm monitoring is divided into collaborative robot arm and industrial robot arm monitoring. Through the DT dashboard, we can remotely monitor the basic operating data of the collaborative robot (Fig. A6-1(a)), such as on/off, coordinates, coordinate trajectory, joint angles, and clamping get other signals; advanced signals can monitor the status of the collaboration robotic arm during the process of running the script (Fig. A6-1(b)), such as whether the control of the robotic arm is obtained, whether the script is operating normally and whether the script operation is stopped, and errors and other related signals. In addition, we can also know the operating status of the robot arm's central control box (Fig. A6-1(c)), such as temperature, power, current and other data, which can ensure the normal operation of the machine and allow the administrator to respond to any occurrences of the robot arm. In case of emergencies, contingency measures can be taken immediately. The same is true for the industrial robot arm, as shown in Fig. A6-2.

The AOI monitoring is divided into signal monitoring for detecting light sources (light source A, light source B, light source C) and tray (X-axis displacement, Y-axis displacement, rotation), as shown in Fig. A6-3. Through DT technology, an AOI DT model is constructed and the physical AOI equipment is monitored using the AOI DT model to achieve virtual and real integration.

The conveyor monitoring is mainly to monitor the operating status of the conveyor, such as power on or off. Through DT technology, a DT model of the conveyor is constructed, and the DT model of the conveyor is used to monitor the physical conveyor equipment to achieve virtual and real integration, as shown in Fig. A6-4.

There are six indicators in the HRC scenario of the DT dashboard, which are the environment in which the personnel is located, production

productivity, heart rate, skin conductance, fatigue level and stress level. Based on literature review, this study designed five levels of fatigue and four levels of stress.

Remotely monitor the operator's physiological status through wearable devices, such as heart rate and skin conductance. Through these signals, the operator's fatigue and stress level in the workplace can be further estimated, allowing managers to know in real-time regarding the current working status of the operator to reduce the chance of industrial safety accidents. In addition, the situation room can also be used to know the extent to which the operator's working status affects work efficiency, as shown in Fig. 11.

Moreover, we can also trace the historical changes of six indicators of personnel (the personnel's environment, productivity, heart rate, skin conductance, fatigue level, and stress level), and clearly understand the period during which personnel are prone to abnormal status. For example, heart rate is too low or too high, fatigue level is increased, skin conductance is too low or too high, stress level is increased, and production efficiency is too low, as shown in Fig. A6-5.

HRC field monitoring is supplemented by real-time on-site images to conduct field monitoring of personnel's working environment. Real-time on-site images can help managers understand the actual situation of the personnel's work site and the types of projects currently being implemented, as shown in Fig. 12.

The IP camera monitoring is combined with the camera system of heterogeneous platforms to transmit video streams through wired networks to achieve remote monitoring and recording. The interface operation in the DT dashboard can replace the original system interface operation. In terms of IP camera settings, there are seven functional operations, namely power on, power off, return to original screen size, image focus, zoom in 300, zoom in 600 and zoom in camera image ratio, as shown in Fig. A4-6(a). By enlarging the camera image ratio, managers can more clearly view the real-time on-site conditions, as shown in Fig. A6-6(b) and A6-6(c).

The current dashboard system is compared with the proposed DT dashboard system of this study, as shown in Table 1. Currently, the equipment dashboard only has the function of monitoring the status of the equipment. The DT dashboard proposed in this study supports the real-time operating status of equipment and has multiple equipment monitoring functions. In addition to basic equipment monitoring, the proposed DT dashboard also adds environment monitoring, personnel monitoring and scheduling monitoring.



Fig. 9. Proposed DT dashboard – Home.



Fig. 10. Proposed DT dashboard – Environment monitoring.

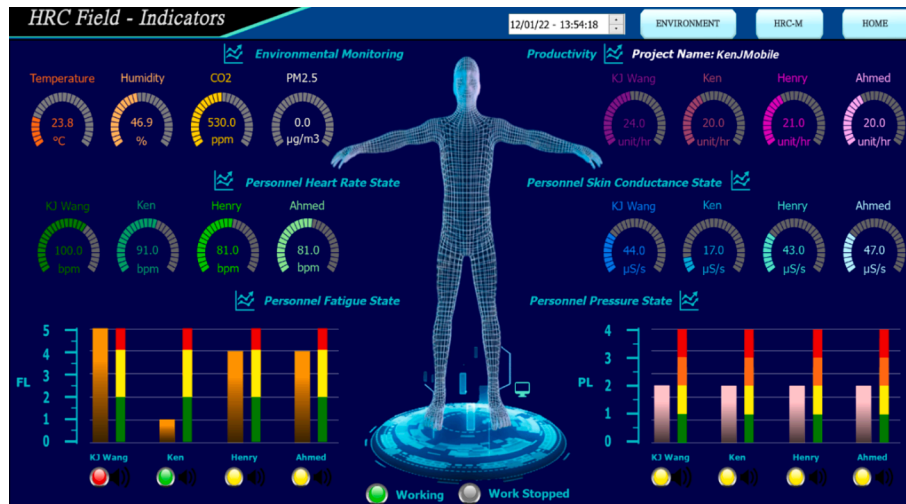


Fig. 11. Proposed DT dashboard – HRC.

5. Discussion on managerial and technological implications of the proposed dashboard framework

Conventional monitoring systems typically offer only data visualization of machines, which lacks a comprehensive understanding of personnel status and immediate feedback. This limitation hinders managers' ability to promptly detect potential safety risks and implement rapid response measures.

The proposed dashboard framework for the GPU card assembly line offers several advantages/innovations over existing dashboards in terms of productivity, quality, accident reduction, and more. Here is a breakdown of how it can be more beneficial: (i) Real-time monitoring and alerts: The proposed dashboard framework provides real-time monitoring of various parameters along the assembly line. This setting allows supervisors and operators to immediately identify any issues or bottlenecks, enabling prompt corrective actions to be taken. By addressing issues promptly, downtime can be minimized, thus enhancing productivity. (ii) Data integration and analysis: Differing from existing dashboard systems that may only provide data visualization, the proposed dashboard integrates data from multiple agents/resources and performs advanced analytics, particularly in scheduling. In addition, predictive analytics to forecast potential equipment failures

and/or quality issues, enabling proactive maintenance and process optimization of the assembly line. By leveraging data effectively, the proposed dashboard framework can help improve overall productivity and product quality. (iii) Quality control features: The proposed dashboard framework includes features specifically designed to monitor and improve GPU card product quality. This situation involves real-time inspection data from vision systems or sensors integrated into the assembly line. By continuously monitoring quality metrics, the proposed dashboard framework can identify deviations from specifications and trigger alerts for immediate action, reducing the likelihood of defective products reaching the market.

In terms of human aspects, the proposed dashboard framework contributes to (i) Safety monitoring and accident prevention: The proposed dashboard framework also includes operator safety/fatigue monitoring features aimed at reducing accidents and ensuring worker safety. It monitors human merits/signals to identify potential safety hazards and track employee adherence to safety protocols. By promoting a safer working environment and preventing accidents, the proposed dashboard framework contributes to both worker well-being and operational efficiency. (ii) Customization and user-friendly interface: A key advantage of the proposed dashboard framework is its customized user-friendly interface. Supervisors and operators are able to customize the

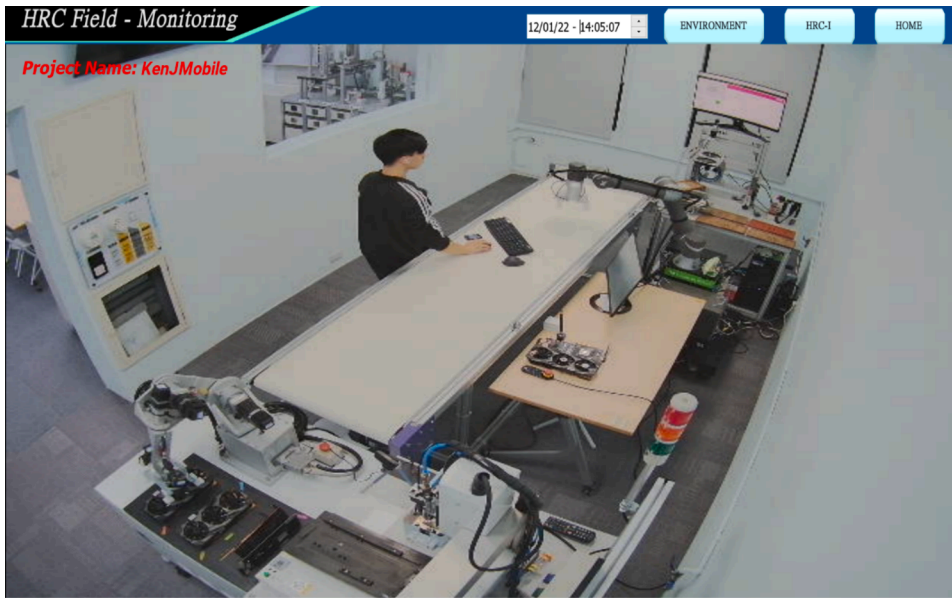


Fig. 12. Proposed DT dashboard – HRC scene real-time image.

Table 1
Comparison of current dashboard system and the proposed DT system.

	Current System	Proposed DT based System
Monitoring content	Equipment focused	• Equipment • Production line • Environment • Personnel
Information velocity	Limited	Real-time
Operations excellence	No	Order scheduling
Communication and connection	Internal Equipment	• Internal Equipment • Modbus TCP/IP • Database connection
Data display and storage	Embedded HMI	• Embedded HMI • CSV file • Database
Decision support	Limited	Real-time

dashboard to display the metrics most relevant to their roles and responsibilities. Additionally, intuitive visualization and navigation features make it easier for users to interpret data and derive actionable insights, leading to more informed decision-making and improved productivity.

6. Conclusions

This study has proposed a novel DT-based dashboard system. By taking the flexible GPU card assembly line as an example, the proposed dashboard system has integrated multiple DT modules, and each DT can communicate through flexible protocols, realizing system operations, scalability, flexibility, and resilience, helping managers solve complex workshop control challenges. In particular, such DT based dashboard framework assists operators for monitoring their status in a real-time situation, in addition to equipment management and production scheduling.

By adopting JMobile to develop personalized interfaces and integrate various data modules from heterogeneous platforms, the proposed dashboard system enhances user experience and transparency of device information, allowing managers to monitor equipment and personal status in real time and in distance of the operator’s physiological status through wearable devices, such as monitoring the person’s heart rate to determine the level of fatigue and monitoring the person’s skin conductance to determine the level of stress, further enhancing the

system scalability from machines to human operators. Integrating human information into the DT based dashboard opens up a way to enhance the application of human-machine collaboration scenarios.

The contribution of this study is reflected in three aspects. First, a DT-based dashboard system is developed to address the complexity of managing a flexible GPU card assembly line. Through digital twin technology, multiple DT modules are integrated to create scene monitoring and applications for real-time monitoring, flexible operation, and enhanced human-machine collaboration. Secondly, a communication protocol is designed that enables each DT to interact and operate flexibly, thereby achieving efficient data exchange and system coordination. This protocol provides the basis for flexible operation and simplified communication throughout the assembly line, addressing the limitations of existing dashboard systems. Third, the proposed DT dashboard framework provides a clear visualization of the relationship between physical devices and virtual models, improving transparency and facilitating informed decision-making by managers. Through the integration of JMobile’s personalized interface and the preliminary design of the personnel status monitoring function, the system’s capabilities are further enhanced, providing an overall solution for monitoring equipment status, environmental conditions, and operator health.

Despite these advancements, our study faces several limitations. The proprietary nature of equipment parameter design restricts modifications by third-party platforms, while certain equipment may prohibit data transmission and parameter writing. In future research and development, the personnel monitoring function can be further optimized, predictive functions can be built, and expansion and updating of equipment compatibility for communication.

CRediT authorship contribution statement

Kung-Jeng Wang: Writing – review & editing, Supervision, Methodology, Funding acquisition, Conceptualization. **Tsung-Lun Lee:** Writing – original draft, Software, Methodology, Investigation, Formal analysis, Data curation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Consent to participate

This manuscript is the author's original work and has not been published nor has it been submitted simultaneously elsewhere.

Consent for publication

All authors have checked the manuscript and have agreed to the submission.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.cie.2024.110491>.

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